WearUS: A Unified Framework for Wearable A-Mode Ultrasound Pretraining and Downstream Transfer

Thirteenth Weekly Report

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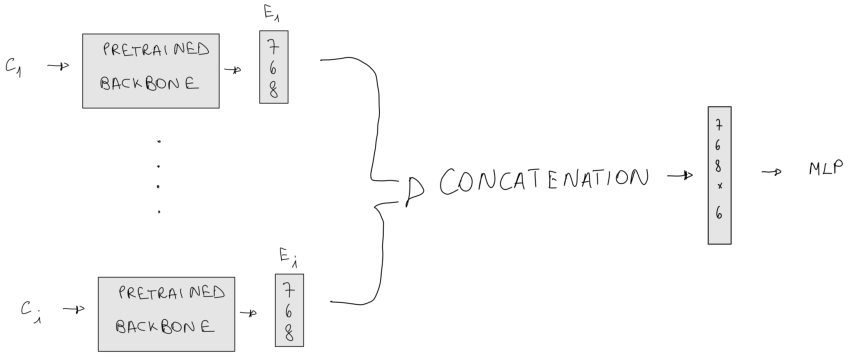
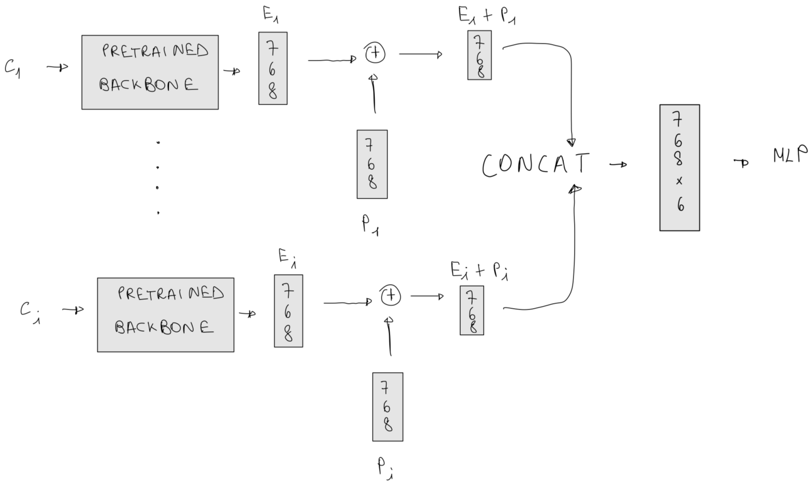
concat

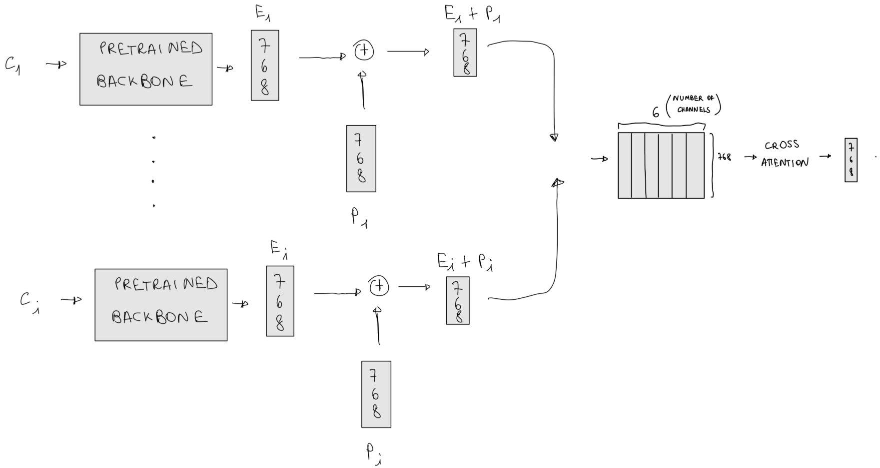


C_1
PRETRAINED
BACKBONE
E_1
768
C_i
PRETRAINED
BACKBONE
E_i
768
CONCATENATION
768 x 6
MLP

posenc


C_1
PRETRAINED
BACKBONE
E_1
768
$E_1 + P_1$
768
768
P_1
CONCAT
768 x 6
MLP
C_i
PRETRAINED
BACKBONE
E_i
768
$E_i + P_i$
768
768
P_i

Channel Cross attention



Results in intra-session testing mode

#	Experiment	Head / Fusion	Gradual Freezing	Test Acc (mean $\pm$ std)
1	concat	concat	no	**70.60% $\pm$ 7.24%**
2	concat_channelShuffle	concat (+ channel_shuffle)	no	60.74% $\pm$ 13.91%
3	posenc	posenc	no	63.40% $\pm$ 15.36%
4	concat_gradualFreezing	concat	yes	69.73% $\pm$ 6.98%
5	posenc_gradualFreezing	posenc	yes	68.79% $\pm$ 7.33%
6	crossattn (num_queries=1)	crossattention	no	55.60% $\pm$ 17.79%
7	crossattn (num_queries=1)	crossattention	yes	56.95% $\pm$ 22.47%
8	crossattn (num_queries=6)	crossattention	yes	58.15% $\pm$ 21.23%

TSNE vs UMAP

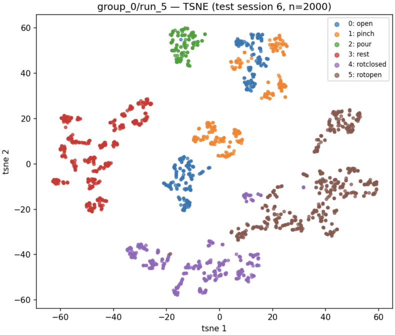


Figure: T-SNE Projection

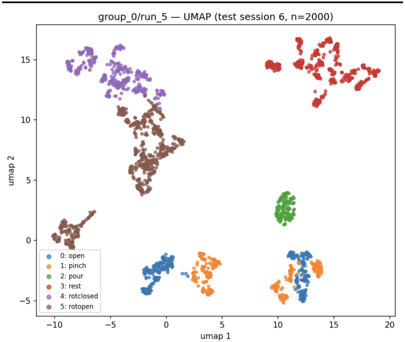
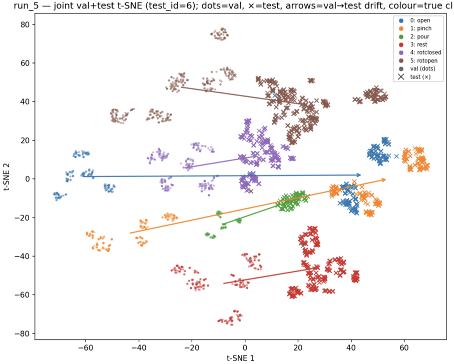
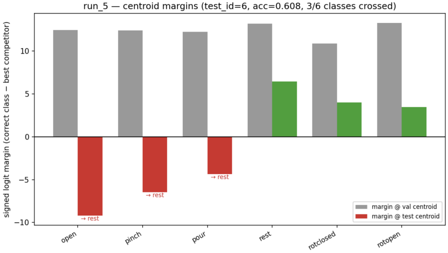


Figure: UMAP Projection

Please also note that the foundational model in Linear Probing, when tested in random mode (trained and tested across all sessions), reaches **98% accuracy** on the test set → the classes are linearly separable in zero-shot conditions.

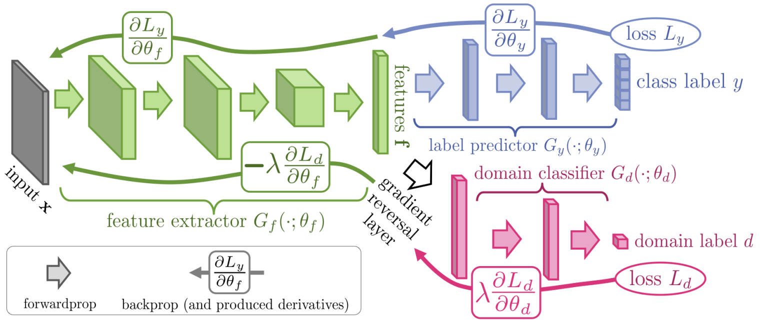
inter-session shift at test-time





Some possible solutions: DANN

Domain-Adversarial Training of Neural Networks - Journal of Machine Learning Research 17 (2016)



Some possible solutions: MSTN

Learning Semantic Representations for Unsupervised Domain Adaptation (ICML 2018)

- It is essentially a DANN where a penalty term is added to the loss to align the **class centroids** between source and target domains (in our case, the different sessions).

Loss Formulation

The total loss is formulated as:

$$\mathcal{L}(\mathcal{X}_S, \mathcal{Y}_S, \mathcal{X}_T) = \mathcal{L}_C(\mathcal{X}_S, \mathcal{Y}_S) + \lambda\mathcal{L}_{DC}(\mathcal{X}_S, \mathcal{X}_T) + \gamma\mathcal{L}_{SM}(\mathcal{X}_S, \mathcal{Y}_S, \mathcal{X}_T)$$

where $\mathcal{L}_C$ is the classification loss, $\mathcal{L}_{DC}$ the adversarial domain loss, and $\mathcal{L}_{SM}$ the semantic matching loss.

ULTRAPRO results

Method	A_1-S_1	A_1-S_2	A_1-S_3	A_2-S_1	A_2-S_2	A_2-S_3	A_3-S_1	A_3-S_2	A_3-S_3	A_4-S_1	A_4-S_2	A_4-S_3	Mean
CNN paper 2022	0.760	0.827	0.764	0.829	0.764	0.813	0.673	0.613	0.629	0.869	0.844	0.800	76.54%
Our foundational model	0.802	0.812	0.778	0.783	0.790	0.767	0.810	0.747	0.760	0.912	0.910	0.888	**81.33%**
New paper 2026	—	—	—	—	—	—	—	—	—	—	—	—	77.42%

Note: Our model on the patient A1 achieves **22% of intra-session test accuracy***. The authors of the paper did not perform inter-session tests, so this is not a directly comparable value.*

References:

- **Original paper:** *Simultaneous Prediction of Wrist and Hand Motions via Wearable Ultrasound Sensing for Natural Control of Hand Prostheses*
- **New paper:** *Parameter-Efficient Deep Learning for Ultrasound-Based Human-Machine Interfaces*